

The continuous-coefficient Jensen equation: A note on vestigial regularity hypotheses

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Abstract. Jensen’s inequality says that for a concave function G and a random variable ξ on an interval, $\mathbb{E}[G(\xi)] \leq G(\mathbb{E}[\xi])$. A natural question: if equality holds for every two-point distribution—if $p G(u_1) + (1 - p) G(u_2) = G(p u_1 + (1 - p) u_2)$ for all u_1, u_2 in the interval and all weights $p \in [0, 1]$ —must G be affine?

The answer is yes. The proof is one line: substitute $u_1 = M$, $u_2 = 0$, $p = v/M$ to read $G(v)$ directly off the straight line through the endpoints. No regularity hypothesis on G —continuity, measurability, monotonicity, or boundedness—is needed.

The result contrasts sharply with what happens when the weights p are restricted to rationals. The $p = \frac{1}{2}$ specialization is equivalent to Cauchy’s additive equation, which admits wildly pathological non-affine solutions (Hamel, 1905) without some regularity hypothesis. Classical analysis from Cauchy (1821) to Ostrowski (1929) cataloged the minimal tameness conditions—continuity, measurability, boundedness on a set of positive measure—that rule the pathology out. These conditions became standard baggage in functional-equation arguments, carried reflexively even where the full continuous-coefficient equation is in play and no such baggage is needed.

This note makes the asymmetry explicit. We catalog which classical regularity hypotheses become vestigial under the continuous-coefficient equation, explain structurally why (the Hamel pathology lives at irrational p , exactly where the continuous-coefficient equation tests G), and survey three application areas—expected-utility theory, the axiomatic characterization of Shannon entropy, and surrogate-loss calibration—where the same vestigial hypotheses recur.

Keywords: Jensen equation, Cauchy equation, Hamel basis, functional equation, regularity hypothesis

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1. INTRODUCTION. One of the most serviceable inequalities in mathematics—Jensen’s inequality, $\mathbb{E}[G(\xi)] \leq G(\mathbb{E}[\xi])$ for concave G —raises an immediate follow-up: when does equality hold for *every* two-point distribution supported on a given interval? The equation expressing universal two-point equality,

$$p G(u_1) + (1 - p) G(u_2) = G(p u_1 + (1 - p) u_2) \quad (u_1, u_2 \in I, p \in [0, 1]), \quad (\star)$$

imposed on a function $G: I \rightarrow \mathbb{R}$ on a real interval I , has at least three forms in classical analysis. These forms have been steadily conflated for over a century, with consequences for both the foundational theory and the modern applied literature where the equation surfaces.

Three forms of the equation. The *discrete-coefficient* Jensen equation,

$$G\left(\frac{u_1 + u_2}{2}\right) = \frac{G(u_1) + G(u_2)}{2}, \quad (J_2)$$

is the $p = \frac{1}{2}$ specialization of $(\star)$. Setting $f(x) := G(x) - G(0)$, the equation (J_2) is equivalent on a translate of I to Cauchy’s additive equation $f(x + y) = f(x) +$

$f(y)$. Cauchy's equation inherits a full apparatus of pathological solutions: without measurability, monotonicity, boundedness on a set of positive Lebesgue measure, or another regularity hypothesis, the equation admits non-affine solutions constructed via a *Hamel basis*—a basis of $\mathbb{R}$ as a $\mathbb{Q}$ -vector space, whose existence requires the axiom of choice. The classical pathological-solution apparatus is the work of Hamel [1]; the regularity hypotheses sufficient to recover affineness were cataloged by Cauchy [2], Darboux [3], Sierpiński [4], Steinhaus [5], and Ostrowski [6].

The *rational-coefficient* Jensen equation,

$$pG(u_1) + (1-p)G(u_2) = G(pu_1 + (1-p)u_2) \\ (u_1, u_2 \in I, \quad p \in [0, 1] \cap \mathbb{Q}), \quad (J_{\mathbb{Q}})$$

strengthens (J_2) but does not escape its pathology. Cauchy additivity entails $\mathbb{Q}$ -homogeneity— $f(qx) = qf(x)$ for every $q \in \mathbb{Q}$ and $x \in \mathbb{R}$ —by an elementary induction on positive integers, sign reversal for negatives, and the inverse-of-multiplication argument for $1/n$. Combining additivity with $\mathbb{Q}$ -homogeneity returns $(J_{\mathbb{Q}})$ for f , hence for G . So $(J_{\mathbb{Q}})$, (J_2) , and Cauchy's equation share the same solution class up to constants, and all three inherit the Hamel-basis pathology.

The *continuous-coefficient* form $(\star)$, by contrast, is fundamentally different. The strengthening from $\mathbb{Q}$ to $\mathbb{R}$ in the range of p is *genuine*. As we will see, $(\star)$ does not admit any Hamel pathology: the pathology cannot survive at irrational p , and the continuous-coefficient equation tests G at every irrational. This is the subject of the paper.

The main result.

Theorem 1. *Let $M > 0$ and let $G: [0, M] \rightarrow \mathbb{R}$ satisfy $(\star)$ for every $u_1, u_2 \in [0, M]$ and every $p \in [0, 1]$. Then G is affine on $[0, M]$:*

$$G(v) = \frac{G(M) - G(0)}{M} v + G(0), \quad v \in [0, M].$$

No regularity hypothesis on G (continuity, measurability, monotonicity, boundedness) is required.

The proof is one line: a substitution into $(\star)$ that pins $G(v)$ on the straight line through the endpoints $(0, G(0))$ and $(M, G(M))$. We give it in Section 2 and use the remainder of the paper to articulate what it means.

The historical detour and the structural moral. Theorem 1 is a folklore result; see, e.g., Aczél's [7, §2.1.4]. It is *not* a new mathematical result. The contribution of the present paper is in three parts:

- *The explicit dictionary* (Section 3) of classical regularity hypotheses on G —continuity, measurability, monotonicity, boundedness on a set of positive measure, and boundedness on the interval—each of which is required to force affineness for (J_2) but becomes *vestigial* under $(\star)$, together with the structural mechanism (the Hamel pathology lives at irrational p , exactly where $(\star)$'s continuous coefficient closes the door).
- *The strict-minimum hypothesis* (Section 4)—the equation $(\star)$ at a single endpoint configuration already suffices.

- *The recurrence pattern* (Section 5)—the trap of invoking unnecessary Cauchy–Hamel regularity recurs whenever a derivation pushes Jensen’s inequality to saturation across a wide class of two-point distributions. We survey three example application areas where this pattern arises.

The classical detour through Hamel and the Polish school answered a perfectly natural question—*what regularity hypothesis suffices for $(J_2) \Rightarrow \text{affine}$?*—but a *different* question from the one that matters for many modern applications. The applied derivation does not produce (J_2) or $(J_{\mathbb{Q}})$; it produces $(\star)$ as a saturated identity, and the one-line substitution of Section 2 is the right closure tool. The dictionary of Section 3 makes explicit which classical regularity hypotheses become vestigial in this transition.

Terminology. Throughout the paper we refer to the substitution $u_1 = M$, $u_2 = 0$, $p = v/M$ in $(\star)$ —by which $G(v)$ is read off the straight line through the endpoints $(0, G(0))$ and $(M, G(M))$ —as the *endpoint substitution*. The substitution is standard folklore in the functional-equations literature (cf. [7, §2.1.4]); the descriptive label is our convenience, adopted for brevity in repeated reference rather than as a claim of established terminology.

2. THE RESULT AND ITS PROOF.

Proof of Theorem 1. Fix $v \in [0, M]$. Set

$$u_1 := M, \quad u_2 := 0, \quad p := \frac{v}{M}.$$

The constraints of $(\star)$ are satisfied: $u_1 = M \in [0, M]$, $u_2 = 0 \in [0, M]$, and $p = v/M \in [0, 1]$ since $v \in [0, M]$. Substituting into $(\star)$,

$$\frac{v}{M} G(M) + \left(1 - \frac{v}{M}\right) G(0) = G\left(\frac{v}{M} \cdot M + \left(1 - \frac{v}{M}\right) \cdot 0\right) = G(v).$$

Rearranging yields $G(v) = G(0) + (G(M) - G(0))v/M$. ■

The endpoint substitution pins $G(v)$ on the straight line from $(0, G(0))$ to $(M, G(M))$ for every $v \in [0, M]$. Since this determines $G(v)$ at every v , the function G is fully determined by its values at the two endpoints—it is exactly the affine function joining them.

Corollary 2. *Under the hypotheses of Theorem 1, G is in particular continuous, monotone, locally Lipschitz, absolutely continuous, and Lebesgue measurable on $[0, M]$.*

Proof. Every affine function $G(v) = av + b$ on a bounded interval is differentiable with constant derivative $G' = a$, hence continuous; non-decreasing if $a \geq 0$ and non-increasing if $a \leq 0$; Lipschitz with constant $|a|$; absolutely continuous; and Borel-measurable, hence Lebesgue-measurable. ■

The order of inference matters and is worth stating explicitly. Under $(\star)$, the regularity properties of G are the *conclusion*, not the hypothesis. Each regularity hypothesis one might be tempted to assume—continuity, monotonicity, boundedness, measurability—is automatically satisfied by any solution of $(\star)$. The continuous-coefficient equation supplies its own regularity from within; no external regularity hypothesis is needed.

This is what makes the continuous-coefficient case structurally different from the discrete-coefficient case (J_2) , where some regularity hypothesis *is* required to rule out the Hamel pathology. We now examine the difference in detail.

3. THE DICTIONARY OF VESTIGIAL REGULARITY HYPOTHESES.

The dictionary. We place side-by-side the discrete-coefficient equation (J_2) and the continuous-coefficient equation $(\star)$, and ask which regularity hypotheses on G are *required* to force affineness for each. Table 1 lists five classical hypotheses, followed by a no-hypothesis row at the bottom. The first column names the hypothesis; the second records its status for (J_2) , with the standard historical attribution; the third records its status for $(\star)$. (For a modern textbook treatment of column 2 — the regularity catalog for (J_2) — we refer the reader to Kuczma [8, §13.2].)

Hypothesis on G	For $(J_2) \Rightarrow \text{affine?}$	For $(\star) \Rightarrow \text{affine?}$
Continuity on I: G is continuous at every $v \in I$.	Yes, suffices (Cauchy [2]).	No —it is the <i>conclusion</i> (Corollary 2).
Measurability on I: G is Lebesgue- or Borel-measurable on I .	Yes, suffices (Sierpiński [4]).	No —same.
Monotonicity on I: G is non-decreasing or non-increasing on I .	Yes, suffices (Darboux [3]).	No —same.
Boundedness on a set of positive measure: G is bounded on some Lebesgue-measurable $E \subseteq I$ with $ E > 0$.	Yes, suffices (Steinhaus [5]).	No —same.
Boundedness on I: G is bounded on all of I .	Yes, suffices (Ostrowski [6]).	No —same.
None: no regularity hypothesis at all.	Insufficient —Hamel-basis pathology [1].	Sufficient —Theorem 1.

Table 1. Regularity hypotheses on G . Column 2 records which hypotheses are classically required to force affineness for the discrete-coefficient equation (J_2) . Column 3 records their status under the continuous-coefficient equation $(\star)$.

The bottom row is the point of the table. **No regularity hypothesis** suffices for (J_2) : there exists a non-affine, non-measurable, unbounded, additive function on $\mathbb{R}$ —the classical Hamel-basis pathology [1]—that supplies a non-affine solution of (J_2) . Conversely, **no regularity hypothesis** is *needed* for $(\star)$: Theorem 1’s endpoint substitution closes the proof at the level of the bare functional equation.

The five non-bottom rows tell a complementary story. For (J_2) , *some* hypothesis from these five is required; otherwise the Hamel pathology survives. For $(\star)$, *none* of these is required—and yet each is automatically satisfied by any solution (Corollary 2). Each hypothesis is, in the dictionary’s terminology, *vestigial*: classically required for (J_2) , classically expected for $(\star)$, but in fact unnecessary for $(\star)$.

The structural mechanism. Why does the Hamel pathology fail to apply to $(\star)$? The mechanism is direct and illuminating.

A Hamel-basis pathological solution G of Cauchy's equation on $\mathbb{R}$ is $\mathbb{Q}$ -linear: $G(0) = 0$ and $G(qx) = qG(x)$ for every $q \in \mathbb{Q}$ and $x \in \mathbb{R}$. By design, G is *not* $\mathbb{R}$ -linear: there exists an irrational $r \in \mathbb{R}$ and an $x \in \mathbb{R}$ such that $G(rx) \neq rG(x)$.

The straight-line identity in $(\star)$ at the configuration $u_1 = M, u_2 = 0$ reads

$$pG(M) + (1-p)G(0) = G(pM). \quad (\star_0)$$

For a Hamel-pathological G (so $G(0) = 0$), $(\star_0)$ becomes $pG(M) = G(pM)$ —exactly the $\mathbb{R}$ -linearity assertion for G at the pair (M, p) . By $\mathbb{Q}$ -linearity of G , this holds for every $p \in \mathbb{Q}$. By the failure of $\mathbb{R}$ -linearity, it *fails* at some irrational p . So the Hamel-pathological G satisfies $(J_{\mathbb{Q}})$ (every rational coefficient is OK) but violates $(\star)$ at every irrational p where the $\mathbb{R}$ -linearity defect appears.

The structural point is one sentence:

The Hamel pathology lives at irrational p ; the continuous-coefficient equation $(\star)$ fore-closes on it there.

The classical program from Cauchy (1821) through Ostrowski (1929) can be read, in hindsight, as an investigation of what regularity hypothesis can substitute for the *unavailable* constraint at irrational p , once it became clear—after Hamel's 1905 construction—that the rational-coefficient version of the equation does not by itself force affineness. With the continuous-coefficient version, the irrational-coefficient constraint is *available*, and the entire dictionary collapses.

4. VARIANTS AND LIMITS.

The strict-minimum hypothesis. The proof of Theorem 1 uses $(\star)$ at exactly one configuration: $u_1 = M, u_2 = 0$, with $p \in [0, 1]$ ranging freely. The full force of $(\star)$ —for every pair (u_1, u_2) —is not consumed. We record the strict minimum.

Theorem 3. *Let $M > 0$ and let $G: [0, M] \rightarrow \mathbb{R}$ satisfy*

$$pG(M) + (1-p)G(0) = G(pM) \quad \text{for all } p \in [0, 1].$$

Then $G(v) = G(0) + (G(M) - G(0))v/M$ on $[0, M]$.

Proof. Set $p := v/M$; then $p \in [0, 1]$ since $v \in [0, M]$. The hypothesis gives

$$pG(M) + (1-p)G(0) = G(pM) = G(v),$$

so $G(v) = G(0) + p(G(M) - G(0)) = G(0) + (G(M) - G(0))v/M$. ■

Theorem 3 is *in principle* weaker than Theorem 1: the one-configuration hypothesis does not a priori imply the full $(\star)$. Under the conclusion (affineness) both hold; in practice an author who derives $(\star)$ from a richer setup (e.g., a Jensen-equality identity in a Bayes-risk computation; see Section 5) typically obtains $(\star)$ in full force. The narrower Theorem 3 is the right reference when one wants to know exactly how much of the equation the proof actually consumes.

Convex domains in higher dimensions. The endpoint substitution extends to any convex subset of a real vector space.

Theorem 4. *Let V be a real vector space, let $C \subseteq V$ be a convex set, and let $G: C \rightarrow \mathbb{R}$ satisfy*

$$pG(x_1) + (1-p)G(x_2) = G(px_1 + (1-p)x_2) \quad (x_1, x_2 \in C, p \in [0, 1]).$$

Then there is a linear functional $a: \text{span}(C - C) \rightarrow \mathbb{R}$ and a constant $b \in \mathbb{R}$ such that $G(x) = a(x - x_0) + b$ on C for any fixed $x_0 \in C$. Equivalently, G is affine on C .

Proof sketch. Apply Theorem 1 along every line segment in C : for each direction v with $x_0 + v \in C$, the map $\lambda \mapsto G(x_0 + \lambda v)$ is affine in $\lambda \in [0, 1]$. Hence $G(x_0 + v) - G(x_0)$ depends linearly on v in each direction; an inductive argument on the dimension of the affine span of the points involved promotes this to a global linear functional a on $\text{span}(C - C)$. The bookkeeping is standard and is given in [9]. ■

The structurally essential step is the one-dimensional endpoint substitution of Theorem 1. The higher-dimensional lifting is bookkeeping.

The rational-coefficient version retains the pathology. The strengthening from rational to continuous coefficients in $(\star)$ is doing genuine work: the rational-coefficient version $(J_{\mathbb{Q}})$ retains the Hamel pathology.

Proposition 5 (Folklore; cf. [8, §§5.2 and 13.2]). *There exists $G: [0, 1] \rightarrow \mathbb{R}$ satisfying $(J_{\mathbb{Q}})$ (hence (J_2)) on $[0, 1]$ that is not affine.*

Construction. Choose a Hamel basis H of $\mathbb{R}$ over $\mathbb{Q}$ containing 1; such a basis exists by Zorn's lemma applied to the partially ordered set of $\mathbb{Q}$ -linearly independent subsets of $\mathbb{R}$. Since H has cardinality $\mathfrak{c}$ and $\mathbb{Q} \cup \{1\}$ is countable, H contains an irrational $h \in (0, 1)$. Define a $\mathbb{Q}$ -linear map $\ell: \mathbb{R} \rightarrow \mathbb{R}$ by setting $\ell(1) = 0$, $\ell(h) = 1$, and extending $\mathbb{Q}$ -linearly on the remaining basis elements.

By construction, ℓ is $\mathbb{Q}$ -linear: $\ell(x + y) = \ell(x) + \ell(y)$ (Cauchy's equation) and $\ell(qx) = q\ell(x)$ for all $q \in \mathbb{Q}$ ($\mathbb{Q}$ -homogeneity). Set $G := \ell|_{[0, 1]}$; then G satisfies $(J_{\mathbb{Q}})$ on $[0, 1]$ because $\mathbb{Q}$ -rational convex combinations of points of $[0, 1]$ remain in $[0, 1]$ and ℓ respects $\mathbb{Q}$ -linear combinations.

Explicit witness that G is not affine: $G(h) = \ell(h) = 1 \neq 0$, while $G(q) = \ell(q) = q\ell(1) = 0$ for every rational $q \in [0, 1]$. Any affine map $A: [0, 1] \rightarrow \mathbb{R}$ agreeing with G on $\mathbb{Q} \cap [0, 1]$ would satisfy $A(0) = 0$ and $A(q) = 0$ for all rational q —forcing $A \equiv 0$, contradicting $G(h) = 1$. ■

Proposition 5 sharpens the dictionary: $(J_{\mathbb{Q}})$ on $[0, 1]$ admits non-affine solutions; only the strengthening to the continuous-coefficient $(\star)$ retires the regularity-hypothesis requirement.

5. WHERE THE TRAP RECURS: JENSEN SATURATION AS A STRUCTURAL PATTERN. The endpoint substitution of Section 2 is folkloric — Aczél's [7, §2.1.4] articulates the underlying observation, and it is standard in the functional-equations community. Why, then, does this paper exist?

Because the explicit packaging matters. The functional-equations community has internalized the substitution; the applied communities that produce the continuous-coefficient equation as a byproduct of Jensen saturation have not. The classical regularity hypotheses — continuity, measurability, and boundedness in particular — keep

reappearing in applied derivations of $(\star)$ outside the functional-equations community. The trap is not a one-off oversight; it has a structural source. This section identifies the source and surveys three application areas where it surfaces.

The structural source: saturated Jensen. The recurrence is predictable. Whenever a derivation in any field arrives at an identity of the form

$$\mathbb{E}[g(\xi)] = g(\mathbb{E}[\xi]) \quad (\xi \text{ a random variable on } I, g: I \rightarrow \mathbb{R}), \quad (\dagger)$$

for a class of random variables ξ wide enough that the marginal $\mathbb{E}[\xi]$ can be any point of I and the support of ξ can be any two-point subset of I with any pair of masses $(p, 1 - p)$, $p \in [0, 1]$, the identity is exactly $(\star)$ with $g = G$ and ξ supported on $\{u_1, u_2\}$ with masses $\{p, 1 - p\}$. Such an identity arises whenever **Jensen's inequality is pushed to saturation**—equality is required in $\mathbb{E}[g(\xi)] \leq g(\mathbb{E}[\xi])$ for concave g across the full two-point family.

Jensen's inequality is the central tool in dozens of fields. In *most* of these fields, Jensen is used as an inequality with explicit slack (tracked in convex analysis via biconjugates and Fenchel duality; in calibration theory via the ψ -transform of Bartlett, Jordan, and McAuliffe [10]; in entropy via convexity arguments). Such derivations never produce $(\star)$ as a saturated identity.

In *some* fields, a derivation produces $(\dagger)$ as a saturated identity: the slack is forced to zero by a structural property of the setup (e.g., the class of ξ is rich enough to exhaust the inequality's freedom). At that moment $(\star)$ appears, and the endpoint substitution is the natural closure. An author unfamiliar with the substitution will reach for the classical Cauchy–Hamel toolkit and find that the regularity hypothesis is unnecessary only after the fact.

Application: expected-utility representation theorems. In **expected-utility theory** in the von Neumann–Morgenstern tradition [11], the 1944 axiomatization of preferences over lotteries—sharpened by Herstein and Milnor [12]—derives a utility functional U on a space of lotteries $\mathcal{L}$ that is *linear in probability*:

$$U(pL_1 + (1 - p)L_2) = pU(L_1) + (1 - p)U(L_2) \quad (L_1, L_2 \in \mathcal{L}, p \in [0, 1]).$$

This is exactly $(\star)$ with U as G and lotteries as points. The classical closure of the linearity step uses the *Archimedean axiom*: for any three lotteries $L_1 \succ L_2 \succ L_3$, there exist $p, q \in (0, 1)$ such that $pL_1 + (1 - p)L_3 \succ L_2 \succ qL_1 + (1 - q)L_3$. The Archimedean axiom forces continuity-in-probability of preferences, which in turn forces continuity of U .

But the endpoint substitution of Theorem 1 *also* closes the linearity step, without any continuity hypothesis. The substitution is the algebraic alternative to the Archimedean axiom. Even where the Archimedean axiom is invoked (and rightly so, as it carries other preference-theoretic content), the closure of the linearity equation $(\star)$ itself does not require it.

Application: Shannon entropy's axiomatic characterization. In the **axiomatic characterization of Shannon entropy** in the tradition of Khinchin [13] and Faddeev [14], Shannon's entropy function $H(p_1, \dots, p_n) = -\sum_i p_i \log p_i$ is characterized by three axioms: continuity, maximum at the uniform distribution, and a *recursivity axiom* that relates the entropy of a joint distribution to a marginal entropy plus a conditional.

In the Khinchin–Faddeev recursivity-axiom analysis, the recursivity axiom reduces, in intermediate steps, to a system of functional equations on auxiliary functions g, h

on $[0, 1]$. Several of these intermediate equations are of $(\star)$ -form for the auxiliary functions, and the closure step uses continuity (the first Khinchin axiom). The endpoint substitution of Theorem 1 is the alternative closure that does not require continuity as a hypothesis; in Aczél–Dhombres [9] the substitution is part of the standard toolkit.

Application: surrogate calibration on the resolution axis. In **surrogate calibration** in statistical decision theory, the foundational tradition — Bartlett, Jordan, and McAuliffe [10] and its extensions by Tewari and Bartlett [15], Steinwart [16], and Reid and Williamson [17, 18] — derives its results via convex analysis (biconjugates, ψ -transforms, Fenchel duality) and consequently tracks Jensen as an *inequality* with explicit slack. Such derivations never produce $(\star)$ as a saturated identity.

A different framing produces $(\star)$ directly. If one passes from the canonical loss-axis formulation to a *resolution-axis* framing — parametrizing risk by the partition that aggregates the feature space rather than by the surrogate loss — a two-cell partition computation forces $(\star)$ on the function G that expresses the partition Bayes risk in terms of the cell mass $p \in [0, 1]$ on an atomless probability space. The full continuous range of cell masses produces the equation in its continuous-coefficient form $(\star)$, and the endpoint substitution closes the linearity step without any regularity hypothesis on G . As elsewhere, an author who has internalized the Cauchy–Hamel toolkit may add a defensive boundedness or measurability hypothesis here; under $(\star)$ that hypothesis is vestigial.

The lesson generalizes. Any derivation that saturates Jensen across a wide class of two-point distributions should be expected to produce $(\star)$ and to benefit from Theorem 1’s one-line closure.

An invitation. The three application areas above are not exhaustive. We invite readers who have encountered $(\star)$ in their own work—in mathematical economics, information theory, statistical decision theory, physics, or elsewhere—and who have invoked an unnecessary Cauchy–Hamel regularity hypothesis to extend the catalog. The structural pattern of saturated Jensen and the endpoint substitution as its natural closure is, we believe, recurrent enough to merit explicit articulation.

6. CONCLUDING REMARKS. Theorem 1 is, mathematically, one line. The dictionary of Section 3 is a table. The structural argument of Section 3 is one sentence: *the Hamel pathology lives at irrational p ; the continuous-coefficient equation $(\star)$ forecloses on it there*. Why does this material deserve an article in the MONTHLY?

Two reasons. First, the classical detour through Hamel and the Polish school produced *foundational* results of classical analysis — and the dictionary of Section 3 articulates a precise sense in which the detour answered an adjacent question: not “what is the minimum hypothesis for $(\star) \Rightarrow \text{affine}$?” (Theorem 1’s answer: nothing), but “what is the minimum hypothesis for $(J_2) \Rightarrow \text{affine}$?” (Polish school’s answer: any of a dictionary of tameness conditions). Both questions are interesting; their answers are different.

Second, the recurrence pattern of Section 5 is, to our knowledge, not articulated explicitly elsewhere in the functional-equations literature. The pattern — saturated Jensen produces $(\star)$, and the endpoint substitution closes it — is recognizable in a number of modern application contexts (utility theory, entropy, calibration), and the explicit articulation may save other authors from carrying vestigial regularity hypotheses in their own work.

The endpoint substitution itself is the kind of one-line proof that, once seen, is hard to unseen — but that nonetheless takes some scaffolding to make visible. The present

paper is the scaffolding.

A note on provenance. The present observation reached the author through an unconventional route: a *formalization project* on a separate topic, in which the lemma “ G satisfies $(\star)$ on $[0, M]$ implies G is affine” was initially stated with a defensive boundedness hypothesis on G , in deference to the Cauchy–Hamel literature. When the formal proof was written out, the boundedness hypothesis turned out to be unused: the endpoint substitution closed the proof on its own. A literature search then confirmed that the substitution is folklore in functional equations (cf. [7, §2.1.4]), and the present paper grew out of the attempt to explain, to the author’s own satisfaction, why the hypothesis had felt necessary in the first place. The lesson is general: a defensive regularity hypothesis is not free, and a formalization project occasionally surfaces such hypotheses precisely because the formal system requires every clause of the statement to be used.

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REFERENCES

- [1] Hamel G. Eine Basis aller Zahlen und die unstetigen Lösungen der Funktionalgleichung $f(x + y) = f(x) + f(y)$. Math Ann. 1905;60:459-62.
- [2] Cauchy AL. Cours d’analyse de l’École royale polytechnique. Première partie: Analyse algébrique. Paris: Imprimerie royale; 1821.
- [3] Darboux G. Mémoire sur les fonctions discontinues. Ann Sci École Norm Sup (2). 1875;4:57-112.
- [4] Sierpiński W. Sur les fonctions convexes mesurables. Fund Math. 1920;1:125-9.
- [5] Steinhaus H. Sur les distances des points dans les ensembles de mesure positive. Fund Math. 1920;1:93-104.
- [6] Ostrowski AM. Über die Funktionalgleichung der Exponentialfunktion und verwandte Funktionalgleichungen. Jahresber Deutsch Math-Verein. 1929;38:54-62.
- [7] Aczél J. Lectures on Functional Equations and Their Applications. vol. 19 of Mathematics in Science and Engineering. New York: Academic Press; 1966.
- [8] Kuczma M. An Introduction to the Theory of Functional Equations and Inequalities. Cauchy’s Equation and Jensen’s Inequality. 2nd ed. Basel: Birkhäuser; 2009. Second edition edited by A. Gilányi.
- [9] Aczél J, Dhombres J. Functional Equations in Several Variables. vol. 31 of Encyclopedia of Mathematics and its Applications. Cambridge: Cambridge University Press; 1989.
- [10] Bartlett PL, Jordan MI, McAuliffe JD. Convexity, classification, and risk bounds. J Amer Statist Assoc. 2006;101(473):138-56.
- [11] von Neumann J, Morgenstern O. Theory of Games and Economic Behavior. Princeton: Princeton University Press; 1944.
- [12] Herstein IN, Milnor J. An axiomatic approach to measurable utility. Econometrica. 1953;21:291-7.
- [13] Khinchin AI. Mathematical Foundations of Information Theory. New York: Dover; 1957.
- [14] Faddeev DK. Zum Begriff der Entropie einer endlichen Wahrscheinlichkeitsschemas. In: Arbeiten zur Informationstheorie. vol. 1. Berlin: Deutscher Verlag der Wissenschaften; 1957. p. 85-90.
- [15] Tewari A, Bartlett PL. On the consistency of multiclass classification methods. J Mach Learn Res. 2007;8:1007-25.
- [16] Steinwart I. How to compare different loss functions and their risks. Constr Approx. 2007;26(2):225-87.

- [17] Reid MD, Williamson RC. Composite binary losses. *J Mach Learn Res.* 2010;11:2387-422.
- [18] Reid MD, Williamson RC. Information, divergence and risk for binary experiments. *J Mach Learn Res.* 2011;12:731-817.